

# Physics A (H156, H556)

## Unified - Newtons laws of motion and momentum

### KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

**OCR supplied materials:**

Additional resources may be supplied with this paper.

**Other materials required:**

- Pencil
- Ruler (cm/mm)

**Duration: Not set**

## INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

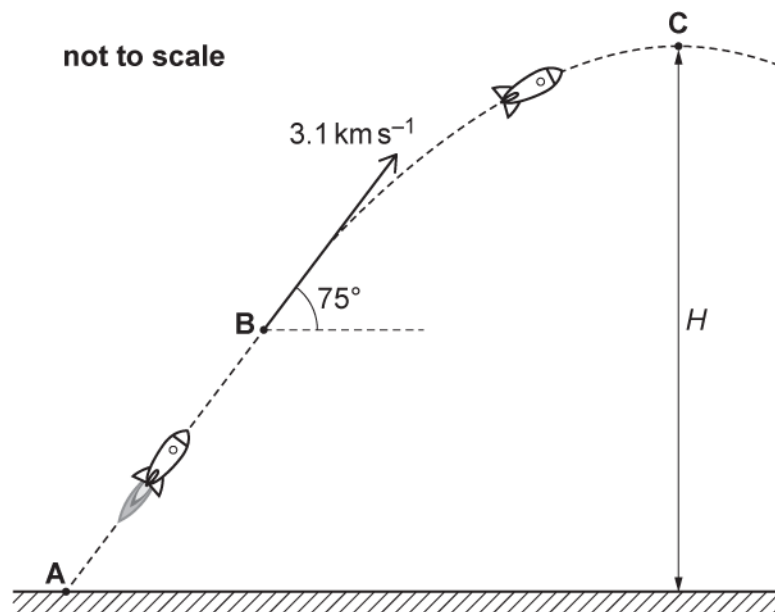
## INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [ ] at the end of each question or part question.
- The total number of marks for this paper is **32**.
- The total number of marks may take into account some 'either/or' question choices.

1 A scientist uses a rocket to study the Earth's atmosphere.

The scientist launches the rocket from rest at the point A at  $t = 0$  seconds. The force produced by the rocket's engine causes it to accelerate.

At  $t = 50$  seconds, the rocket's engine no longer produce an accelerating force as all of the fuel has been used. The rocket has reached point B. Its velocity is now  $3.1 \text{ km s}^{-1}$  at an angle of  $75^\circ$  to the horizontal.



The rocket engine works by expelling hot gas backwards.

Explain, using Newton's laws of motion, how the engine causes the rocket to accelerate between  $t = 0$  and  $t = 50$  s.

-----

-----

-----

-----

-----

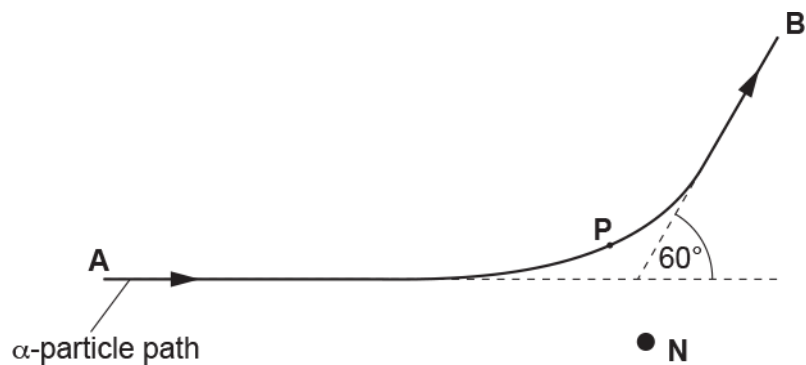
-----

[3]

- 2(a) A beam of  $\alpha$ -particles is incident on a thin gold foil. Most  $\alpha$ -particles pass straight through the foil. A few are deflected by gold nuclei.

The diagram shows the path of one  $\alpha$ -particle which passes close to a gold nucleus **N** in the foil. The  $\alpha$ -particle is deflected through an angle of  $60^\circ$  as it travels from **A** to **B**.

**P** marks its position of closest approach to the gold nucleus.



The initial kinetic energy of each  $\alpha$ -particle is 5.0 MeV.

Show that the magnitude of the initial momentum of each  $\alpha$ -particle is about  $10^{-19} \text{ kg m s}^{-1}$ .

Take the mass of the  $\alpha$ -particle to be  $6.6 \times 10^{-27} \text{ kg}$ .

[3]

- (b) The **magnitude** of the final momentum of the  $\alpha$ -particle at **B** is equal to its initial value at **A**.

The gold nucleus **N** is initially at rest. During the passage of the  $\alpha$ -particle from **A** to **B**, no other forces act on the two particles.

In the following questions label any relevant angles.

- i. Draw two vectors in the spaces below to represent the initial momentum and the final momentum of the  $\alpha$ -particle.

initial momentum at **A**

final momentum at **B**

[2]

- ii. Draw a vector in the space below to represent the momentum of the nucleus **N** when the  $\alpha$ -particle reaches **B**.

Explain how you determined this momentum.

-----

-----

-----

[2]

- 3(a) A stationary uranium-238 nucleus  $\left(^{238}_{92}\text{U}\right)$  decays into a nucleus of thorium-234 by emitting an alpha-particle.

The chemical symbol for thorium is Th. Write a nuclear equation for this decay.

[2]

- (b) The mass of the uranium nucleus is  $4.0 \times 10^{-25}$  kg. After the decay the thorium nucleus has a speed of  $2.4 \times 10^5 \text{ m s}^{-1}$ .

Calculate the kinetic energy, in MeV, of the alpha-particle.

kinetic energy = \_\_\_\_\_ MeV [4]

- (c) The uranium-238  $\left(^{238}_{92}\text{U}\right)$  nucleus starts the decay chain which ends with a nucleus of lead-206  $\left(^{206}_{82}\text{Pb}\right)$ .  
Show that 14 particles are emitted during this decay chain. Explain your reasoning.

[3]

Fig. 3.1 shows a simple representation of a hydrogen iodide molecule. It consists of two ions  ${}^1_1\text{H}^+$  and  ${}^{127}_{53}\text{I}^-$ , held together by electric forces.



Fig. 3.1

Fig. 3.2 shows a simple mechanical model of the molecule consisting of two unequal masses connected by a spring of force constant  $k$  and negligible mass. The ions oscillate in simple harmonic motion when disturbed.

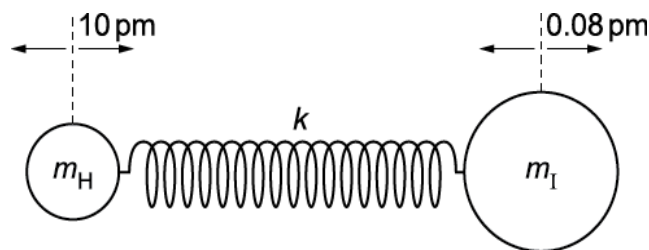


Fig. 3.2

- i. The approximate acceleration  $a$  of the hydrogen ion, mass  $m_{\text{H}}$ , is given by the equation

$$a = -\left(\frac{k}{m_{\text{H}}}\right)x$$

where  $k$  is the force constant of the spring and  $x$  is the displacement of the ion.

The ions oscillate with a frequency of  $6.6 \times 10^{13}$  Hz. The mass  $m_{\text{H}}$  is  $1.7 \times 10^{-27}$  kg.

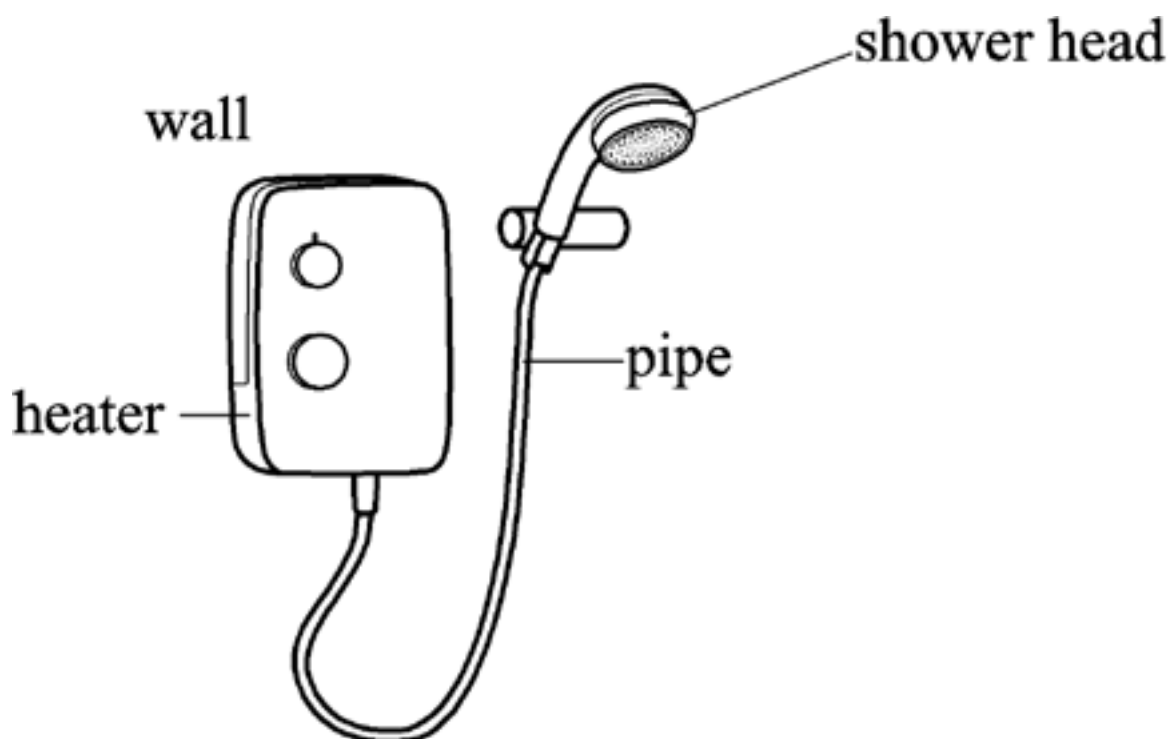
Show that the value of  $k$  is about  $300 \text{ N m}^{-1}$ .

[3]

- ii. Use Newton's laws of motion and a requirement for simple harmonic motion to explain why the amplitude of oscillation of the iodine ion, mass  $m_{\text{I}}$ , is about  $0.08 \text{ pm}$  when the amplitude of oscillation of the hydrogen ion is about  $10 \text{ pm}$ .

[4]

- 5 This question is about the operation of an electrically powered shower designed by an electrical firm.



**Fig.1.1**

Water moves at constant speed through a pipe of cross-sectional area  $7.5 \times 10^{-7} \text{ m}^2$  to a shower head shown in Fig. 1.1. The maximum mass of water which flows per second is  $0.070 \text{ kg s}^{-1}$ .

- i. Show that the maximum speed of the water in the pipe is about  $0.9 \text{ m s}^{-1}$ .

density of water =  $1000 \text{ kg m}^{-3}$

[2]

- ii. The total cross-sectional area of the holes in the shower head is one quarter that of the pipe. Calculate the maximum speed of the water as it leaves the shower head.

maximum speed = \_\_\_\_\_  $\text{m s}^{-1}$  [1]

- iii. Calculate the magnitude of the force caused by the accelerating water on the shower head.

force = \_\_\_\_\_ N [2]

- iv. Draw on to Fig. 1.1 the direction of the force in (iii).

[1]

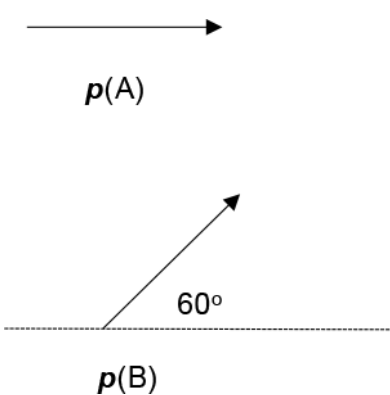
**END OF QUESTION PAPER**



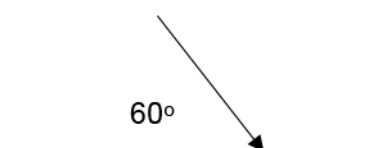
### Mark Scheme

| Question |  |  | Answer/Indicative content                                                                                                                                                                                                                                                            | Marks                         | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        |  |  | <p>the rocket exerts a (backwards) force on the (hot) gas</p> <p>(so) the gas exerts a (forwards) force on the rocket</p> <p>(this forwards) force must be greater than the weight of the rocket / there is a resultant force (which causes acceleration / a change in momentum)</p> | <p>B1</p> <p>B1</p> <p>B1</p> | <p><b>Ignore</b> incorrect numbering or general statements of Newton's Laws throughout</p> <p><b>Allow</b> engine for rocket</p> <p><b>Allow</b> (burning) fuel for gas but <b>not</b> air</p> <p>Note that 'rocket expels gas backwards' is in the stem and so gains no credit</p> <p><b>Allow</b> engine for rocket</p> <p><b>Allow</b> gas pushes the rocket forwards</p> <p><b>Allow</b> net or unbalanced for resultant</p> <p><u><b>Examiner's Comments</b></u></p> <p>It was expected that candidates would use, rather than simply state, Newton's laws of motion in order to explain why the rocket accelerates. Most candidates were able to state, 'for every action there is an equal and opposite reaction' or even 'if A exerts a force on B, then B exerts an equal and opposite force on A'. However, few extended that to 'if the rocket exerts a force on the gas (backwards) then the gas will exert a force on the rocket (forwards)'. Even fewer went on to explain that, in order for the rocket to accelerate, this force must be greater than the weight of the rocket in order for there to be a resultant force (forwards).</p> |
|          |  |  | <b>Total</b>                                                                                                                                                                                                                                                                         | <b>3</b>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

# Mark Scheme

| Question |   |   | Answer/Indicative content                                                                                                                                                                                                                                                                                                                                                                       | Marks                         | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2        | a |   | <p>(k.e. =) <math>E = 5.0 \times 10^6 \times 1.6 \times 10^{-19}</math></p> <p><math>v = \sqrt{2E/m}</math> or <math>= \sqrt{(2 \times 8.0 \times 10^{-13} / 6.6 \times 10^{-27})} = 1.6 \times 10^7 \text{ (ms}^{-1}\text{)}</math></p> <p><math>p (= mv) = 6.6 \times 10^{-27} \times 1.6 \times 10^7</math><br/>giving <math>p = 1.1 \times 10^{-19} \text{ (kg m s}^{-1}\text{)}</math></p> | <p>C1</p> <p>C1</p> <p>A1</p> | <p><math>E = 8(.0) \times 10^{-13} \text{ J}</math></p> <p>or (<math>E = p^2/2m</math> so) <math>p = \sqrt{2mE}</math><br/> <b>Note:</b> A value of <math>v = 1.6 \times 10^7 \text{ (ms}^{-1}\text{)}</math> automatically scores both C1 marks even if the calculation for <math>E</math> is not shown</p> <p>or <math>p (= \sqrt{2mE}) = \sqrt{(2 \times 6.6 \times 10^{-27} \times 8.0 \times 10^{-13})}</math><br/> giving <math>p = 1.0 \times 10^{-19} \text{ (kg m s}^{-1}\text{)}</math><br/> Full substitution of values must be shown and answer (if calculated) must be correct</p> <p><b><u>Examiner's Comments</u></b></p> <p>This question provided an excellent opportunity for candidates to produce immaculate responses: identify the physics involved, select and write down the correct formula, do the necessary conversion (MeV to J), rearrange the formula, substitute correctly and then write the final response in standard form to a correct number of significant figures. Some of the common errors were:</p> <ul style="list-style-type: none"> <li>• forgetting to convert 5.0 MeV into J</li> <li>• not showing a full substitution of values (which is necessary for a 'show that' question)</li> <li>• not calculating the response to more than 1 s.f. (which is necessary for a 'show that' question).</li> </ul> |
|          | b | i | <p>Example (not to scale):</p>  <p><math>p(A)</math></p> <p><math>60^\circ</math></p> <p><math>p(B)</math></p>                                                                                                                                                                                               | <p>B1</p> <p>B1</p>           | <p>horizontal arrow (judge by eye), in the direction shown</p> <p>arrow drawn at an angle of <math>60^\circ</math> to the horizontal (angle must be shown), in the direction shown</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

# Mark Scheme

| Question |  |    | Answer/Indicative content                                                                                                                                                                                       | Marks               | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|--|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |  | ii | <p>Example (not to scale):</p>  <p>(Can apply principle of) conservation of momentum (since no external forces are acting)</p> | <p>B1</p> <p>B1</p> | <p>arrow drawn at an angle of <math>60^\circ</math> to the horizontal (angle must be shown), in the direction shown</p> <p><b><u>Examiner's Comments</u></b></p> <p>This was not an easy question but, even so, a good number of candidates did well. The marks were given for the direction (rather than for the magnitude) of the momentum vectors. Some of the common errors were:</p> <ul style="list-style-type: none"> <li>• forgetting to label relevant angles</li> <li>• not using arrows to show direction</li> <li>• drawing a vector triangle without any indication of which arrow was meant to be the final momentum.</li> </ul> |
|          |  |    | Total                                                                                                                                                                                                           | 7                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

# Mark Scheme

| Question |   | Answer/Indicative content                                                                                                                                                                                                                                                                                                                                   | Marks                                   | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3        | a | ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \dots$ ${}^4_2\text{He} \text{ or } {}^4_2\alpha$                                                                                                                                                                                                                                               | <p>B1</p> <p>B1</p>                     | <p>allow proton and / or nucleon number to the right of symbol</p> <p>allow <math>\gamma</math>-photon; zero for any other extra particle</p> <p><b>Examiner's Comments</b><br/>Most candidates made a good start to the paper writing a correct equation for the nuclear decay.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|          | b | $mv = (4.00 - 0.0665) \times 10^{-25} \times 2.40 \times 10^5$ $= 9.44 \times 10^{-20}$ $v = 9.44 \times 10^{-20} / 6.65 \times 10^{-27} = 1.42 \times 10^7$<br>$\text{k.e.} = \frac{1}{2} \times 6.65 \times 10^{-27} \times (1.42 \times 10^7)^2$ $= 6.70 \times 10^{-13} \text{ (J)}$ $6.70 \times 10^{-13} / 1.60 \times 10^{-13} = 4.19 \text{ (MeV)}$ | <p>C1</p> <p>C1</p> <p>A1</p> <p>B1</p> | <p>allow <math>0.07 \times 10^{-25}</math> for <math>\alpha</math>-particle mass</p> <p>max 3 if use 4.00 instead of 3.93 in momentum eq'n</p> <p>allow ratio of masses 234 and 4 or calculations using 234u and 4u</p> <p>allow <math>p^2/2m</math> calculation for k.e.</p> <p>accept 4.0 to 4.2; ecf (calculated value of k.e. in J)/e</p> <p>N.B. the correct answer automatically gains all 4 marks</p> <p><b>Examiner's Comments</b><br/>One mark in this question was reserved for converting units from joule into mega electronvolt. This was the only mark awarded to half of the candidates. Few recognised this to be an isolated system, applying the conservation of momentum to solve the problem. Few appeared to realise that the mass of an alpha particle is given in the Data, Formulae, and Relationships Booklet, calculating it instead by summing the masses of neutrons and protons. The most common incorrect approach was to use the formula <math>E = mc^2</math> or to equate the kinetic energies of the thorium nucleus and alpha particle.</p> |

# Mark Scheme

| Question |   |    | Answer/Indicative content                                                                                                                                               | Marks          | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|---|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | c |    | $\Delta A = 32 = 4n_\alpha$ so $n_\alpha = 8$<br>$\Delta Z = 10 = 2n_\alpha - n_\beta$ so $n_\beta = 6$<br>argument / reasoning given for both $n_\alpha$ and $n_\beta$ | B1<br>B1<br>B1 | allow 8 (decays), i.e no mention of $\alpha$ particles<br>allow $10 - 16 = -6$ ; NOT $14 - 8 = 6$ ; <b>must state</b> $\beta(-)$ particles<br>e.g. change in mass number caused by $\alpha$ decay, change in proton number combination of $\alpha$ and $\beta$<br><br><b>Examiner's Comments</b><br>A significant number had no idea where to start and left the page blank. Of the rest most managed to decide on 8 alpha particles. A minority worked initially with the proton number rather than the nucleon number incorrectly choosing 5. The explanations about the choice of 6 beta particles were often just restricted to equating the numbers correctly rather than giving any description of the transformation of neutrons into protons. |
|          |   |    | <b>Total</b>                                                                                                                                                            | <b>9</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4        |   | i  | $a = 4\pi^2 f^2 \times$                                                                                                                                                 | C1             | condition for SHM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|          |   | i  | so $k = (m4\pi^2 f^2) = 1.7 \times 10^{-27} \times 4 \times 9.87 \times 43.7 \times 10^{26}$                                                                            | B1             | substitution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|          |   | i  | $k = 292 \text{ (N m}^{-1}\text{)}$                                                                                                                                     | A1             | <b>ecf</b> if incorrect mass used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|          |   | ii | (N2 gives) $F_H = m_H a_H$ and $F_I = m_I a_I$                                                                                                                          | B1             | <b>allow</b> total momentum = 0 at all times                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|          |   | ii | (N3 gives) $F_H = F_I$ <i>can be implicit</i>                                                                                                                           | B1             | SHM gives $v = 2\pi f x_{\max}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|          |   | ii | SHM gives $a \propto (-)x$                                                                                                                                              | B1             | so $m_H x_H = m_I x_I$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|          |   | ii | hence $x_H/x_I = a_H/a_I = m_I/m_H = 127$                                                                                                                               | B1             | <b>accept</b> $127 = x_H/x_I \approx 10/0.08 = 125$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|          |   |    | <b>Total</b>                                                                                                                                                            | <b>7</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### Mark Scheme

| Question |  |     | Answer/Indicative content                              | Marks    | Guidance                             |
|----------|--|-----|--------------------------------------------------------|----------|--------------------------------------|
| 5        |  | i   | $\rho = m/V = m/Av$ ; so $m = Apv$                     | C1       |                                      |
|          |  | i   | $7.5 \times 10^{-5} \times 1000 \times v = 0.070$      | A1       |                                      |
|          |  | i   | giving $v = 0.93 \text{ (m s}^{-1}\text{)}$            | A0       |                                      |
|          |  | ii  | $3.7 \text{ (m s}^{-1}\text{)}$                        | A1       | Accept 3.72                          |
|          |  | iii | $F = \Delta(mv)/\Delta t = 0.070 \times (3.72 - 0.93)$ | C1       | ecf (ii)                             |
|          |  | iii | $F = 0.195 \text{ (N)}$                                | A1       | accept 0.19 or 0.2(0)                |
|          |  | iv  | arrow into the shower head perpendicular to its face.  | B1       | award mark for a reasonable attempt. |
|          |  |     | <b>Total</b>                                           | <b>6</b> |                                      |